Appendix. Feature correlation and the effect of redundancy on classification performance

1.1 Scope and method

This appendix analyses the correlations among the 10 common-component features used for lithology identification in the main text, quantifies their redundancy, and tests by ablation whether that redundancy affects the classification performance reported there. The correlation and redundancy diagnostics are computed on the samples that actually enter the classifiers; the ablation follows the protocol of the main text (the same 10 random seeds, a class-stratified 80/20 split, 5-fold grid search on the training set, and the mean over 10 repeats) and changes only the input feature subset.

Of the three feature subsets, F10 keeps all 10 features and is the configuration used in the main text; F8 drops root mean square and variance, which are by definition near-duplicates of the mean and the standard deviation respectively; and F6 additionally drops peak-to-peak and the interquartile range, leaving the standard deviation as the sole representative of the highly correlated dispersion family.

1.2 Correlations among the features

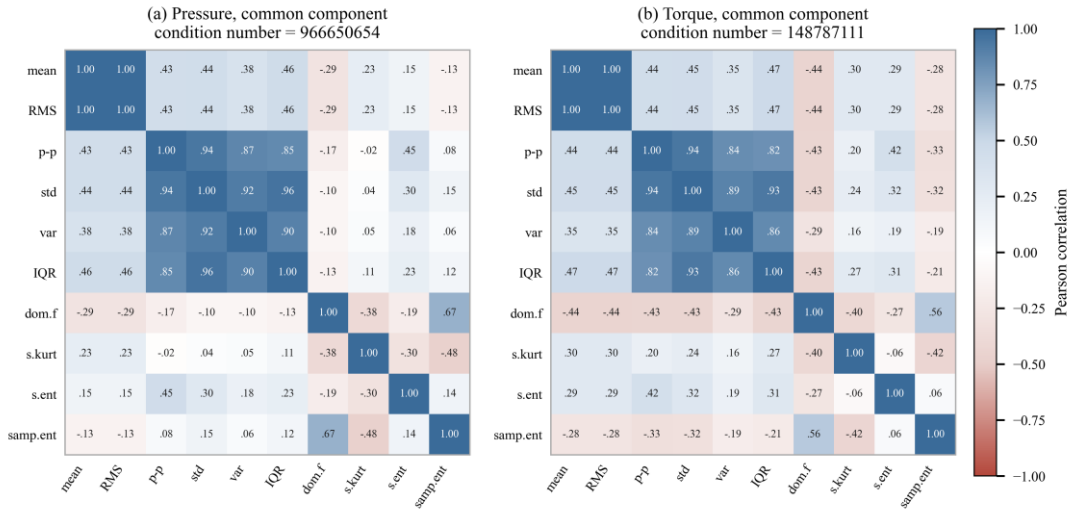


Fig. 1. Pearson correlation matrix of the 10 common-component features.

Table 1. Feature pairs with an absolute correlation coefficient above 0.90.

Signal	Feature A	Feature B	Pearson r
Pressure	Mean	Root mean square	+1.0000
	Standard deviation	Interquartile range	+0.9606
	Peak-to-peak	Standard deviation	+0.9393
	Standard deviation	Variance	+0.9185
Torque	Mean	Root mean square	+1.0000
	Peak-to-peak	Standard deviation	+0.9448
	Standard deviation	Interquartile range	+0.9325

The redundancy is concentrated in two places. The first is the mean and the root mean square, whose correlation coefficient is +1.0000 for both the pressure and the torque signal, so that they are almost perfectly collinear; this arises because the fluctuation amplitude of the common component is far smaller than its absolute level, which reduces the root mean square numerically to the absolute value of the mean. The second is the dispersion family (standard deviation, variance, interquartile

range and peak-to-peak), whose pairwise correlations lie between 0.82 and 0.96 and which represent different measures of the same amplitude dispersion. Correlations among the spectral features, and between them and the first two groups, are generally low, the highest pair being dominant frequency and sample entropy (+0.673 for pressure and +0.563 for torque).

1.3 Quantifying the redundancy

Table 2. Variance inflation factor (VIF) of each feature and the condition number of the correlation matrix.

Feature	Pressure VIF	Torque VIF
Mean	1.07×10^{8}	1.49×10^{7}
Root mean square	1.07×10^{8}	1.49×10^{7}
Peak-to-peak	19.6	17.1
Standard deviation	50.5	38.2
Variance	8.3	6.2
Interquartile range	21.2	14.0
Dominant frequency	2.4	2.1
Spectral kurtosis	1.6	1.4
Spectral entropy	2.2	1.8
Sample entropy	2.6	2.0
Condition number of the correlation matrix	9.67×10^{8}	1.49×10^{8}

The VIFs of the mean and the root mean square are far higher than the rest, consistent with their near-perfect collinearity; those of the dispersion family come next, and those of the spectral features are close to 1. The condition number of the correlation matrix likewise indicates a pronounced ill-conditioned direction in the feature space.

1.4 Effect of the redundancy on classification performance

Table 3. Classification accuracy for the three feature subsets (mean over 10 repeats, %).

Signal	Feature subset	LDA	KNN	SVM-RBF	XGBoost	LightGBM	Random Forest
Pressure	All 10 (10-D)	89.9	96.5	98.5	98.8	99.0	98.7
	Drop 2 (8-D)	89.7	95.5	98.0	98.8	98.9	98.5
	Drop 4 (6-D)	89.5	95.3	98.1	98.9	98.9	98.6
Torque	All 10 (10-D)	84.5	94.4	96.1	96.7	97.1	96.9
	Drop 2 (8-D)	82.7	94.1	96.0	96.6	97.4	96.9
	Drop 4 (6-D)	82.5	93.5	95.1	96.5	96.7	96.3
Pressure + torque	All 10 (20-D)	90.9	96.9	98.1	99.3	99.5	99.3
	Drop 2	89.3	95.9	97.8	99.4	99.5	99.2

	(16-D) Drop 4 (12-D)	89.5	95.0	97.7	99.3	99.5	99.3

Table 4. Macro-averaged F1 for the three feature subsets (mean over 10 repeats, %).

Signal	Feature subset	LDA	KNN	SVM-RBF	XGBoost	LightGBM	Random Forest
Pressure	All 10 (10-D)	73.7	88.6	96.3	96.7	97.3	96.3
	Drop 2 (8-D)	72.1	85.6	94.4	96.6	97.2	94.7
	Drop 4 (6-D)	71.8	85.5	94.8	97.2	96.8	95.3
Torque	All 10 (10-D)	58.2	81.9	90.0	93.8	94.8	94.8
	Drop 2 (8-D)	53.5	80.5	89.9	93.7	95.5	94.6
	Drop 4 (6-D)	53.0	79.3	86.1	93.4	94.2	93.6
Pressure + torque	All 10 (20-D)	79.4	89.7	94.0	98.5	99.0	98.6
	Drop 2 (16-D)	76.0	86.8	92.8	98.6	99.0	98.2
	Drop 4 (12-D)	76.3	84.9	92.3	98.5	98.9	98.5

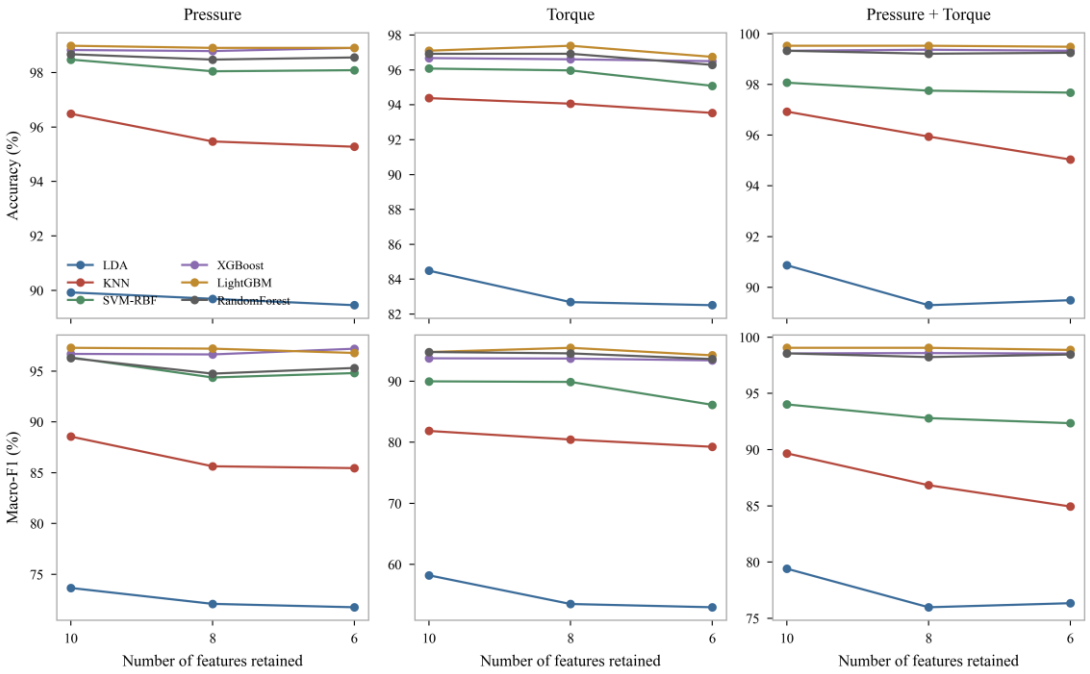


Fig. 2. Classification performance against the number of features retained.

Reducing F10 to F6 cuts the feature dimension by 40%. The highest accuracy of the three signal combinations changes from 99.0%, 97.1% and 99.5% to 98.9%, 96.7% and 99.5%

respectively, the best configuration reported in the main text (pressure + torque) changing by less than 0.1 percentage points. Across all 18 comparisons (3 signal combinations × 6 classifiers) accuracy changes by -2.0 to +0.1 percentage points (median -0.4) and macro-averaged F1 by -5.2 to +0.5 percentage points (median -1.3); of these, only 4 accuracy comparisons and 3 macro-F1 comparisons exceed the repeat-to-repeat standard deviation of the combination concerned.

The distribution of these changes depends clearly on the classifier. The tree ensembles (XGBoost, LightGBM and Random Forest) are almost unaffected, with accuracy changing by -0.6 to +0.1 and macro-averaged F1 by -1.2 to +0.5 percentage points; the losses are concentrated in the linear discriminant, nearest-neighbour and kernel methods. The largest drop in accuracy is for LDA on the torque signal (-2.0 percentage points) and the largest drop in macro-averaged F1 is likewise for LDA on the torque signal (-5.2 percentage points).

Two conclusions follow. First, removing redundant features produced no improvement beyond the repeat-to-repeat fluctuation in any comparison, so the redundancy did not inflate the performance reported in the main text by introducing spurious degrees of freedom. Second, apart from the perfectly collinear pair of the mean and the root mean square, the members of the dispersion family are not interchangeable despite correlations as high as 0.82–0.96: peak-to-peak captures the extreme range, the interquartile range a robust measure of spread, and the standard deviation the second moment, so the three respond to different aspects of the same distribution, and compressing them into a single representative costs models that rely on the covariance structure part of their discriminative information. This also explains why the losses are concentrated in LDA and KNN while the tree ensembles, which split on one feature at a time, are almost unaffected.

At the modelling level, of the six classifiers used in the main text the tree ensembles select one feature at a time when splitting, and the nearest-neighbour and kernel methods act on standardized distances, so none of the three is sensitive to collinearity. The only one affected by collinearity in principle is linear discriminant analysis, and the main text uses the lsqr solver for it with the shrinkage coefficient included in the grid search, a mechanism designed precisely for an ill-conditioned covariance matrix. The influence of redundancy on the best results of the main text is therefore already controlled by design.

1.5 Discriminative contribution of each feature

Table 5. One-way ANOVA F value of each feature (common component, in descending order of the pressure signal).

Feature	Pressure F value	Torque F value
Root mean square	4929.8	4392.8
Mean	4929.5	4391.7
Peak-to-peak	80.3	86.4
Interquartile range	73.0	84.9
Dominant frequency	69.8	105.4
Standard deviation	69.7	81.2
Variance	66.9	66.4
Sample entropy	35.5	57.2
Spectral kurtosis	31.3	54.5
Spectral entropy	19.9	78.9

The ranking by F value shows that the discriminative information lies mainly in the amplitude features and some of the spectral ones, the root mean square having the highest F value of all. The

F values of the mean and the root mean square are almost identical, confirming once more that the two carry the same information.

1.6 Conclusions

There is indeed clear redundancy among the 10 features: the mean and the root mean square are almost perfectly collinear (r = +1.0000), and correlations within the dispersion family lie between 0.82 and 0.96. That redundancy does not, however, threaten the conclusions of the main text. First, removing it produced no improvement beyond the repeat-to-repeat fluctuation in any of the 18 comparisons, so it did not inflate the reported performance. Second, when the features are reduced from 10 to 6 the highest accuracy of the three signal combinations changes by at most 0.4 percentage points, and the best configuration reported in the main text by less than 0.1. Third, the performance losses are concentrated in the linear discriminant and nearest-neighbour methods, whereas the best results of the main text come from tree ensembles, which are insensitive to collinearity. All 10 features are retained in order to characterize the signal completely without presupposing which features matter more; the correlation analysis shows that this choice neither supports nor damages the classification performance reported.